



OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

Software Requirements Specification (SRS)

Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

Software Requirements Specification (SRS)



- A **software requirements specification (SRS)** is a **document** that describes what the **software will do** and **how it will be expected to perform**. It also describes the functionality the product needs to fulfill all stakeholders (business, users) needs.
- A Software requirements specification (SRS) document describes the **intended purpose, requirements, and nature of software/application/project to be developed**.
- To prepare an SRS document, one should need to have a functional knowledge about the project or application, **knowledge of software/hardware/technology** to be used.

-
- **SRS Format:** as name suggests, is complete specification and description of requirements of software that needs to be fulfilled for successful development of software system.
 - These requirements can be functional as well as non-requirements depending upon type of requirements.
 - The interaction between different customers and contractor is done because its necessary to fully understand needs of customers

- In order to form a good SRS, consider the following points:

1. Introduction

- (i) Purpose of this document
- (ii) Scope of this document
- (iii) Overview

2. General description

3. Functional Requirements

4. Interface Requirements

5. Performance Requirements

6. Design Constraints

7. Non-Functional Attributes

8. Preliminary Schedule and Budget

9. Appendices

Depending upon information gathered after interaction, SRS is developed

Introduction :

(i) Purpose of this Document –

At first, **main aim** of why this document is necessary and what's purpose of document is explained and described

(ii) Scope of this document –

In this, **overall working** and main objective of document and what value it will provide to customer is described and explained. It also includes a description of **development cost and time required**.

(iii) **Overview** – In this, description of product is explained. It's simply summary or overall review of product.

2. General description :

In this, general functions of product which includes **objective of user, a user characteristics, features, benefits, about why its importance** is mentioned. It also describes features of user community

3. Functional Requirements :

In this, possible outcome of software system which includes effects due to operation of program is fully explained. All functional requirements which may include calculations, data processing, etc. are placed in a ranked order.

4. Interface Requirements :

In this, software interfaces which mean **how software program communicates** with each of its users either in form of **any language, code, or message** are fully described and explained. Examples can be shared memory, data streams, etc.

5. Performance Requirements :

In this, **how a software system performs desired functions under specific condition** is explained. It also explains required time, required memory, maximum error rate, etc.

6. Design Constraints :

In this, constraints which simply means **limitation or restriction** are specified and explained for design team. Examples may include **use of a particular algorithm, hardware and software limitations**, etc.

7. Non-Functional Attributes :

In this, non-functional attributes are explained that are required by software system for better performance. An example may **include Security, Portability, Reliability, Reusability, Application compatibility, Data integrity, Scalability capacity**, etc.

8. Preliminary Schedule and Budget :

In this, initial version and budget of project plan are explained which include **overall time duration required and overall cost required for development** of project.

9. Appendices :

In this, additional information like **references** from where information is gathered, definitions of some **specific terms**, **acronyms**, **abbreviations**, etc. are given and explained.

Functional Vs Non-functional Requirements

- **Functional requirements** are those which are related to the **technical functionality** of the system.
- **Non-functional requirements** are the requirements that specifies criteria that can be used to judge the operation of a system in particular conditions, rather than specific behaviors.

- **Functional requirements** are the **product features** or **its functions** that must be designed directly for the users and their convenience. They define the functionality of the software, which the software engineers have to develop so that the users could easily perform their tasks up to the business **requirements**
- **Nonfunctional Requirements (NFRs)** define system attributes such as **security, reliability, performance, maintainability, scalability, and usability.**



THANK YOU

Ms. Archana A

Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392